

Cholesterol oxides in Indian ghee: possible cause of unexplained high risk of atherosclerosis in Indian immigrant populations.

Two populations of immigrants to London and to the West Indies from the Indian subcontinent have higher than expected morbidity and mortality from atherosclerosis but do not show the commonly accepted major risk factors. This study investigated the hypothesis that ghee, a clarified butter product prized in Indian cooking, contains cholesterol oxides and could therefore be an important source of dietary exposure to cholesterol oxides and an explanation for the high atherosclerosis risk. Substantial amounts of cholesterol oxides were found in ghee (12.3% of sterols), but not in fresh butter, by thin-layer and high-performance-liquid chromatography. Dietary exposure to cholesterol oxides from ghee may offer a logical explanation for the high frequency of atherosclerotic complications in these Indian populations.